A Forgotten Particle May Explain Why the Universe Exists

From: ScienceDaily Top Technology Mined: 2026-11-02 5 min read

Scientists have proposed a surprising new explanation for why the universe exists at all, and it centers on a mysterious particle that physicists have largely overlooked for decades. Researchers from the University of Southampton and the Indian Institute of Technology Kanpur say this forgotten particle may have helped matter survive after the Big Bang.

The puzzle is simple to state and hard to solve. Scientists believe matter and **antimatter** were made in nearly equal amounts in the early universe. When opposites meet, they destroy each other, so exactly equal amounts would have left almost nothing to build stars, planets or life.

The team's answer involves a heavier relative of the **neutrino**, an extremely lightweight particle that rarely interacts with ordinary matter. This type, called a '**decoupled**' heavy **neutrino**, was long dismissed as too weakly connected to matter. But the researchers calculated that it decays incredibly slowly, so it lived long enough to temporarily dominate the universe's energy as the cosmos cooled after the Big Bang.

That brief dominance could have changed the temperature at which fundamental particles, like electrons and other **leptons**, interacted with their antiparticles with the help of the Higgs boson. That temperature acts like a switch, said study author Dr. Rishav Roshan of the University of Southampton, shaping how the small surplus of matter over **antimatter** built up. "The fact that we are here at all proves that an **asymmetry** between matter and **antimatter** occurred," he said.

The findings were published in Physical Review Letters, a journal of the American Physical Society. The idea is a proposed explanation, not proof, but it shows how something long considered insignificant may have helped decide why the universe contains matter at all.

Word Treasure

antimatter -- The opposite of matter; when they meet, they destroy each other.

neutrino -- A tiny particle that rarely interacts with normal matter.

decoupled -- Only weakly connected to other particles.

asymmetry -- A difference where one side has more than the other.

leptons -- Basic particles, like electrons.

Background you need

Matter is the stuff everything is made of, and antimatter is its opposite.

The Big Bang is the hot, dense beginning of the universe, after which it cooled and expanded.

Neutrinos are extremely light particles that pass through ordinary matter almost without touching it.

The Higgs boson is a particle linked to how other particles get mass.

Break it down

LEAD: A forgotten particle may explain why the universe exists
KEY EVENT: Researchers propose a decoupled heavy neutrino helped matter survive
+- WHO: Researchers from the University of Southampton and the Indian Institute of Technology Kanpur
+- WHAT: A heavier relative of the neutrino
+- PUZZLE: Matter and antimatter were nearly equal, and opposites destroy each other
MECHANISM: Slow decay let it dominate the universe's energy and shift the temperature switch
QUOTE: Dr. Rishav Roshan says being here proves an asymmetry occurred
STATUS: Published in Physical Review Letters as a proposed explanation, not proof

Why it matters

If a particle everyone ignored helped decide whether anything exists at all, it shows how the smallest, quietest things can shape the biggest questions.

Test what you remember

1. What puzzle does the research address?

- (A) Why neutrinos rarely interact
- (B) Why the universe is cooling
- (C) Why the universe contains matter instead of nothing
- (D) Why stars and planets formed

2. What kind of particle is at the center of the idea?

- (A) An ordinary neutrino
- (B) A decoupled heavy neutrino
- (C) A Higgs boson
- (D) An electron

3. What did researchers calculate about the particle's lifetime?

- (A) It decays incredibly slowly
- (B) It decays in seconds
- (C) It never decays
- (D) It decays instantly

4. What did the particle temporarily dominate as the cosmos cooled?

- (A) The speed of light
- (B) The universe's energy
- (C) The Higgs boson
- (D) All antimatter

5. Who is the study author quoted in the article?

- (A) Physical Review Letters
- (B) University of Southampton
- (C) Dr. Rishav Roshan
- (D) American Physical Society

6. Where were the findings published?

- (A) American Physical Society
- (B) Physical Review Letters
- (C) University of Southampton
- (D) The Big Bang

Answer key (try the quiz first!): 1: C 2: B 3: A 4: B 5: C 6: B

Questions to think about

1. Dr. Rishav Roshan: The decoupled neutrino shifted the temperature that shaped how the matter-antimatter imbalance built up.

MY THOUGHTS (at least 20 words):